Masa: AI-Adaptive Mobile App for Sustainable Agriculture

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***Abstract*— Mobile applications have made significant positive contributions to our daily lives in recent years. They have greatly improved practices in critical sectors such as healthcare, education, and agriculture etc. The agricultural sector is under pressure to meet the ever-increasing food demand while needing to deal with lack of agricultural resources (such as water and soil) and climate change issues. There is also a need to figure out how to attract and involve young people in agriculture in order to replace aging farmers. These issues necessitate the development of new, sustainable agricultural solutions. This is the demand that Masa app is attempting to meet. This paper presents the design and implementation of an AI-adaptive mobile app for sustainable agriculture. The app is divided into two sections: the *market section* which creates a platform where buyers are able to connect with the farmers directly and the *resource center* section where the power of AI is used to give guidance to the farmers. It also has learning resources where new entrants into the Agric sector can have first-hand walk-through guides in the Agric field containing all the information they need to venture into farming any product of choice. This will help mitigate some of the challenges faced by people who are new or venturing into the agricultural sector thereby encouraging more people to enter agriculture hence its sustainability. Initial feedback on the application from researchers of human-computer interaction domain show the application's promise in closing the gap in farming knowledge among general people while highlighting some limitations and suggestion that can be considered for improving the application.**

Keywords*— *Mobile App, AI, Adaptive System, Sustainable Agriculture, Persuasive Technology, HCI4D

I. Introduction

There is no single, well-defined end goal in sustainable agriculture. The scientific understanding of what constitutes agricultural sustainability is constantly growing, and it is influenced by current concerns, viewpoints, and beliefs. For instance, agriculture's ability to adapt to climate change was not regarded as a serious concern 30 years ago but is now gaining increasing attention. Furthermore, the specifics of what makes a sustainable system might vary depending on the circumstances (e.g., climate, security of farmers especially in developing countries, commercialization), as well as one cultural and ideological standpoint to the next, leaving the term "sustainable" a contentious concept. As a result, rather than thinking of agricultural systems as either sustainable or unsustainable, think of them as ranging along a continuum from unsustainable to very sustainable.

According to Food and Agriculture Organization of the United Nations, sustainable agriculture dwells on the principle of farming in a way that meets society's current food and textile needs without jeopardizing posterity's ability to meet their own needs [1]. In the past, the subject of sustainable agriculture has been approached from various perspectives in a bid to address challenges surrounding its sustainability, these include agroecology, nature-inclusive agriculture, permaculture, biodynamic agriculture, organic farming, conservation agriculture, regenerative agriculture, carbon farming, climate-smart agriculture, high nature value farming, low external input agriculture, circular agriculture, ecological intensification, and sustainable intensification [2]. This paper tends to approach this subject matter from the viewpoint of creating a platform (mobile app) to encourage and motivate people to venture into agriculture.

Research [3] has shown that one of the issues facing sustainable farming includes lack of interest in the agricultural sector. This is due to the many reasons which can be seen as challenges to upcoming farmers. Some of these challenges amongst others include lack of skill and risk-bearing ability, low productivity due to poor knowledge or experience, and challenges in marketing agricultural products.

Taking the Nigerian farm industry as a case study, Ikuemonisan et al. [4] have shown that Nigeria is currently the largest producer of cassava in the world with an estimated figure of about 59.9 million metric tons. During the 2nd National Commodity Alliance Forum held in Nigeria (2020), the President of Nigeria, President Muhammadu Buhari noted that small holders' farmers especially cassava producers in Nigeria have expressed their frustration and lamentation over the inability to sell their fresh cassava roots [5]. These are some concerns that need to be addressed so that people can be encouraged to go into farming while doing so in a sustainable way.

Following these issues, this paper presents the design, implementation, and informal evaluation of an AI adaptive mobile app for sustainable agriculture called Masa. The app is separated into two sections:

A. The resource center section:

Agrarians must seed the right plant, at the right time, and in the right place in order to produce abundant crops [6]. The ideal location is determined not only by geographic location and climate characteristics but also by soil types. As such, the key to succeeding in farming here is to understand the specific soil type and to grow plants that are best suited to the condition. In this section, we leveraged the power of AI to keep the farmers well informed. Lack of information or delayed information, when combined with other factors such as environmental shocks, results in a significant loss in crop produce, quality, or sale price, and ultimately the farmer suffers [7].

This section provides personalized learning tools where new entrants to the agricultural sector can get a first-hand walk-through guide in the agricultural field with all the knowledge they need to get started with farming. This will help persons who are new to or starting into the agricultural sector overcome some of the problems they will confront, hence reduce the barrier to entry. As a result, more people will be drawn to agriculture, ensuring its sustainability.

B. The market section:

Arbabi et al. [8] indicate that the inefficiency of the agricultural marketing system is one of the major factors limiting agricultural development goals and farmer income in developing countries. This section creates a platform for buyers to connect directly with farmers. One of the major problems in marketing agricultural products is long chain of middlemen [9] [10]. Agricultural goods perhaps, have the longest chain of middlemen; there are a number of intermediaries in the market like the wholesalers, brokers, commission agents, retailers, and so on.

Oguoma et al. [11] established that farmers face high production costs in their efforts to increase output, but they rarely receive fair pricing for their products from middlemen. The real profit goes to the middlemen, who buy farm products for almost nothing and sell them to consumers at exorbitant prices. Because of the marginal profit associated with agriculture, and because the middlemen cart away the majority of the profits; genuine investors have been discouraged from investing in agriculture. But with the advent of Masa app, farmers will be able to sell their crops directly to traders or final consumers while also reducing food prices.

This research will help mitigate some of the challenges faced by people who are new or venturing into the agricultural sector and also encourage more people to venture into agriculture. The societal need for agriculture cannot be over-emphasized hence the need for its sustainability.

The following is a breakdown of the paper's structure: in section II, we go over theories that underpin our AI adaptive mobile app design. Section III details the system's design, development, and implementation, as well as an overview of the app's features and how they can benefit users. In Section IV we discuss the initial feedback on our application that was collected through conducting interviews with researchers from the human-computer interaction domain. Finally, in Section V, we discussed the app's usefulness, drawbacks, and future work.

II. BACKGROUND OF THE STUDY

In this section, the theories that govern our AI adaptive mobile app design will be discussed.

A. Sustainable Agriculture

The concept of sustainable agriculture and rural development was first presented in the Netherlands in 1991 at the United Nations Food and Agriculture Organization (FAO) conference on Agriculture and Environment in Hertogenbosch (den Bosch) [12]. This conference resulted in a significant collection of papers as well as a Declaration and Agenda for Action on Sustainable Agriculture and Rural Development (SARD). This Declaration identified three critical goals for achieving SARD:

- Food security;
- Rural employment and income generation (aimed at eradicating poverty); and
- Natural resource conservation and environmental protection [12].

According to a leading financier of agriculture, World Bank Group, agriculture can help reduce poverty, raise incomes and improve food security. The need for sustainable agriculture is an ever-growing need and cannot be overlooked.

B. Mobile Applications for Sustainable Agriculture

The significance of mobile technologies in our daily activities and interactions cannot be overemphasized. The use of mobile applications has spread to many different sectors in recent years, including education, health, and agriculture, while shaping them for the greater good. Despite the fact that agriculture and mobile app technology appear to be unrelated on the surface, there is substantial evidence that using mobile and cloud-based applications not only addresses these sustainability issues but also creates commercial value for both large agribusiness companies and small-scale farmers.

C. Artificial Intelligence in Mobile Apps

Artificial Intelligence (AI), machine learning, and IoT have propelled apps to a new level, with everything geared toward providing users with actionable and well-informed insights. Research [13] has shown that there are numerous examples where AI already is making an impact on the world and augmenting human capabilities in significant ways and are also capable of tremendous sophistication in analysis and decision making. Machine learning is increasingly being used by app developers to improve the user experience and enhance the usefulness of applications.

D. Mobile Apps Adaptability

Adaptive layouts are about how an app displays the user interface on the device it is being used on. Research of different types of natural and human systems such as done by Alam et al. [14] have taught us that those systems that survive over time usually do so because they are highly resilient, adaptive, and have high diversity.

Resilience is essential because most agroecosystems face conditions (such as climate, pest populations, political contexts, and others) that are highly unpredictable and rarely stable over time. Adaptability on the other hand is an important component of resilience because, while it may not always be possible or desirable for an agroecosystem to return to its original form and function after a disturbance, it may be able to adjust and take on a new form in the face of changing conditions [14] [15].

Studies [16] have shown that application-aware adaptation holds a lot of promise as a way to deal with a mobile client's ever-changing environment. Therefore, it is imperative that mobile applications be designed in such a way that they can dynamically adapt to different platforms to improve user experience.

E. AI-Driven Adaptive Systems

AI-driven adaptive systems are capable of providing more adaptive and efficient information for users than traditional adaptive systems. To achieve this, we apply Deep reinforcement learning (deep RL). Deep RL is a machine learning subfield that incorporates both reinforcement learning (RL) and deep learning (DL). The problem of a computational agent learning to make decisions by trial and error is addressed in RL. Deep RL integrates deep learning into the solution, allowing agents to make decisions based on unstructured input data without the need for manual state-space engineering [17].

III. System Design and Implementation

This section discusses the technologies and tools used in the design and implementation of Masa. Screenshots from different pages of the application are provided in Fig. 2, 3, 4, and 5.

A. Architecture

The architecture and design patterns of an application play a significant role in its development. A layered architecture is used in the system design, which is implemented using the Model-View-ViewModel (MVVM) architecture of Flutter [18]. The MVVM design pattern provides a uniform representation of data while emphasizing adaptability and suitability. The general application layout can be seen in Fig. 1.

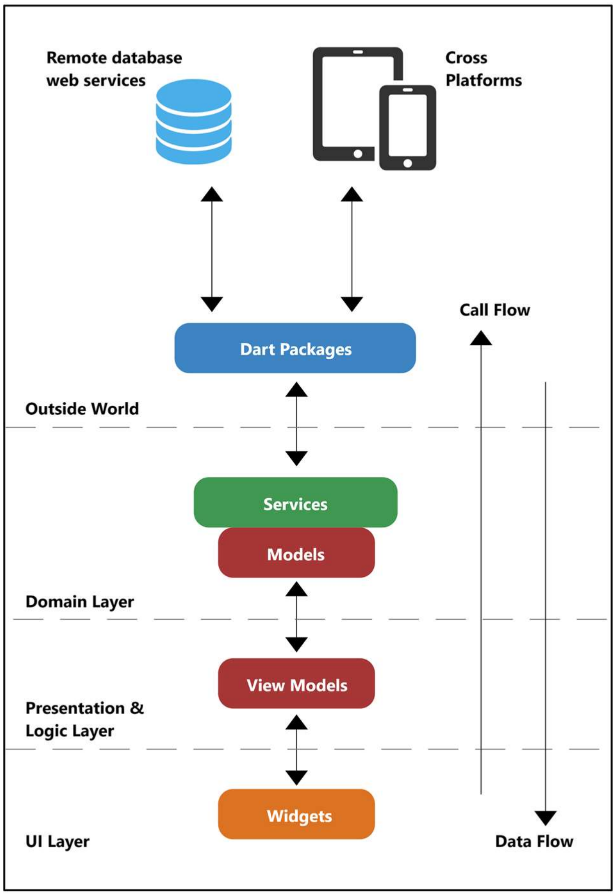

Fig. 1. Flutter Application Layout

B. Frontend Technology

The frontend (mobile interfaces) was implemented using the Flutter framework. Flutter is Google's UI toolkit for building beautiful, natively compiled applications for mobile, web, and desktop from a single codebase. Flutter is powered by Dart, an object-oriented programming language created by Google. Flutter was chosen for this project because of its exceptional adaptability with libraries that are highly scalable.

C. Backend Implementation

The backend functionalities are yet to be developed, and it involves the implementation of a couple of services and frameworks. The backend of this design will be implemented using Google Firebase [19]. Firebase is a Backend-as-a-Service (BaaS) app development platform that offers hosted backend services like a real-time database, cloud storage, authentication, crash reporting, machine learning, remote configuration, and static file hosting. Several Firebase services will be used in this project including Firebase Authentication, Firebase Machine Learning Kit, Cloud Firestore, and Cloud Storage.

Firebase Authentication will be used to register and authenticate users. It also affords us the possibility to register with numerous identity providers such as Facebook, Twitter, Google, and GitHub. This project also will use Cloud Firestore, a NoSQL document database to store, sync, and query data for our mobile app. In addition, Cloud Storage would be used to store and serve user-generated content such as photos uploaded by marketplace sellers quickly and easily. It works with Firebase Authentication to restrict unauthorized access.

To implement our weather forecast feature, we will fetch real-time weather data from the internet using the OpenWeatherMap API call [20]. The OpenWeatherMap API, which is powered by convolutional machine learning, can deliver all of the weather data needed for decision-making for any location in the world. Also, to implement our AI-driven adaptive system, we shall implement deep reinforcement learning algorithm as earlier mentioned in section II. First, we will create our environment using OpenAI Gym, a toolkit for developing and comparing reinforcement learning algorithms. Then a deep learning model using TensorFlow and Keras (API) can be built. Finally, we will pass the built model into Keras RL (a library for Deep Reinforcement Learning with Keras) in order to train our RL model using policy-based learning.

We will make use of the Flutter Geolocator Plugin [21] to ascertain the user's current geographic location and listen for changes. Knowing the exact location of the users is crucial in the app's resource center, as it assists in providing optimal resources by selecting the most appropriate learning material for each user based on their location and matching it with the best crops that can grow in their location.

D. Masa Application Functionality

Masa is an adaptive mobile app aimed at promoting sustainable agriculture. The mobile app makes it easy for people to get started in the agricultural sector. New entrants can get all of the information they need to get started at their fingertips. Also, when it’s time to market their farm produce, farmers can easily overcome the unfair competitive advantage that experienced farmers have because they can quickly find buyers for their product through the app. The app consists of two main sections, the resource center section and the market section.

When users visit the app for the first time, they have to register as seen in Fig. 2, before they can proceed to log in. When they successfully log in, they are taken to the home page as shown in Fig. 3, where they have the option to visit the marketplace or the resource center. The resource center provides learning materials and also leverages the power of AI to analyze climate conditions for the users. On the

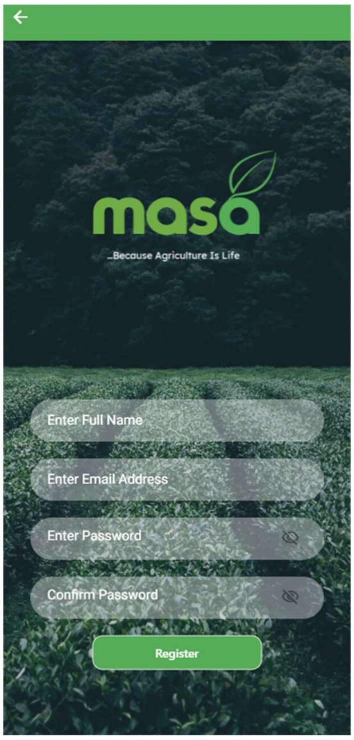

Fig. 2. Registration Screen

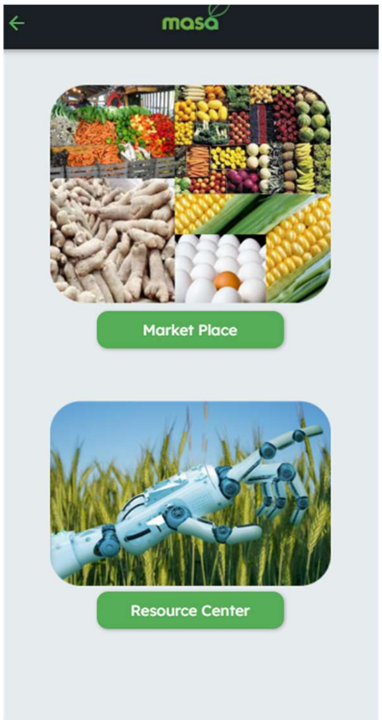

Fig. 3. Home Screen

marketplace, as seen in Fig. 4, farmers can easily post their products for buyers to patronize them.

IV. FEEDBACK ON THE APPLICATION

We conducted informal interviews with 5 computer science researchers from our university lab who focus on the human-computer interaction domain research. The goal of the interviews was to collect some initial feedback on the application's design and features. All the participants were in the later stage of their master's degree, except for one participant who was doing a Ph.D. Among the participants, 3 were male and 2 were female. The interviews were semi-structured, allowing us to pursue interesting responses with further questions. The interviews began with the lead author introducing the application to the participant and walking them through the different sections of the application, after which the second author asked the participants questions regarding the application features. The feedback from the participants is summarized below, where they are identified by their identifiers: P1, P2, P3, P4, and P5.

A. Reception on the Purpose of the Application

All the participants were intrigued by learning about the application and appreciated the idea behind it. Participant P3 commented: "*Idea is really good, we do need it… we need more people to get interested in farming*". All of them also specifically mentioned that being able to directly purchase items from the farmers, eradicating any need for a middleman is one of the best features of the app. Participant P4 shared concerns of dishonesty among the middlemen in selling agricultural produce and stated that excluding them from the picture is an appreciative move. When asked about the most likable section of the app, all the participants stated that they liked the Marketplace section the most. P1 commented: "*We seek for farmers who sell healthy chicken, it's good to be able to contact them*". Few participants also mentioned that the app was organized well and seemed easy to use.

All participants liked the Resource Center's weather forecast feature, while P3 wanted to see detailed forecast than only for 7 days. The quick guides in this section were also commented upon, one of the comments from P4 being "*guides also impressed me a lot*". P1 asked if they could communicate with an AI agent or experts in agriculture to solve problems they are facing in their farming. This was a great feature suggestion for us to consider in improving the app. Few participants asked to have video tutorials in the app in addition to textual guides.

B. Identified Challenges and Drawbacks of the Application

Participants were able to bring out some drawbacks and challenges to the application's functionalities. Most of their concerns revolved around securing the marketplace to prevent malicious sellers and protecting the users' privacy. Participant P2 commented: "*I would like to verify the sellers…rating and comments on the seller could help verify the sellers*". Most users also stated that it might be difficult to attract users to the app, especially, for farmers who are used to traditional methods of farming. P3 commented: "*understanding the resource center might be difficult for farmers who are traditional, they might ignore the guide and sustainable farming altogether and follow their old methods*". P4 raised concerns on relying on artificial intelligence, stating "*AI is prone to errors*" which shows that in addition to AI-driven suggestions on farming, it might be beneficial for the app to have opinions from expert persons which goes in line with the suggestion from P1 as stated above. It was mentioned by P5 that the effectiveness of the guides would rely on how easy it will be to understand them, emphasizing the importance of how well the guides are prepared.

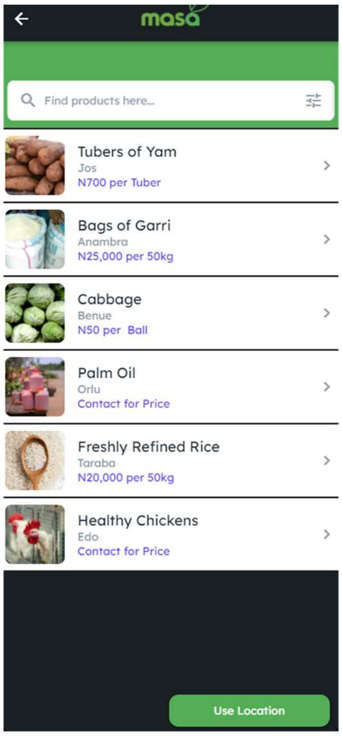

Fig. 4. Market Place Screen

Fig. 5. Resource Center Screen

C. Recommendation on the Application

Participants offered many useful recommendations for the app, from user interface design to adding new features. Participant P4 identified the lack of contrast in the design, suggesting to make the background less noticeable than the foreground of the app. Participant P5 suggested making the Resource Center the main page of the app after logging in and making the marketplace a subsection of the app. Most participants suggested having a detailed description of what sustainable farming is and how it benefits the farmers, as some believed not all users would be aware of sustainable farming. Participant P4 suggested having rewards system on the app to incentivize users who successfully follow the sustainable farming process. P5 recommended having a gradual introduction to sustainable farming for beginners i.e., first introducing them to home gardening, then vertical farming to sustainable farming as it would be easier for beginners to adapt to the process.

The feedback from the participants were very positive, showing the promise of the app in fulfilling its purpose. The identified issues and recommendations are valuable considerations for improving the application. These feedback in an initial stage of the development of the application paved the way in further understanding of our app itself; while helping us plan ahead on the development to avoid mistakes and building a better application overall.

V. Discussion

Agriculture involves a variety of farming activities, ranging from soil preparation to storage and possible marketing of produce [22]. The smooth operation of these activities is critical for crop productivity and resource sustainable practices. Masa as an adaptive app personalizes the users' experience by providing useful contents that are of interest to that particular user. The app employs the deep reinforcement learning algorithm to search the internet for meaningful contents that are relevant to the user. This is done by detecting the user's location and matching it with their agricultural needs. Fig. 5, illustrates an adaptive experience by user Aaditya, from Delhi. Delhi is known to have alluvial soils almost entirely covering the city's landmass [23]. These soils contain a suitable amount of potassium, phosphoric acid, and lime, making them ideal for sugarcane, paddy, wheat, and other cereal and pulse crops [23]. Considering these factors, Masa is able to provide resource materials on crops that have the potential to perform better in Aadiya's location in a bid to optimize yields and minimize water requirements for irrigation. The app considers attributes that affect agricultural activities and stays updated in real-time to adapt to the increasing need of agricultural resources.

Some drawbacks of the app include possible spamming of the marketplace with fake products/wrong pricing by malicious users, which was also one of the main concerns identified by participants in the interviews. Secondly, farmers with improper knowledge of technology might not find it useful and may need proper training before using the app. Moreover, farmers who are residual in remote areas with poor or no internet access may find it difficult to use the app. The guidelines posted on the resource center need to be accurate and verified from appropriate sources so that people are not misguided, especially the new farmers who constitute a major target user group for this application. In addition to these, the identified drawbacks, challenges, and recommendations that surfaced from the interview sessions give us adequate work to consider for the improvement of the Masa app. We hope to continue building the application and run a full-fledged user study to gather more feedback that will help us justify the effectiveness of the application as well as identifying more improvements.

VI. CONCLUSION

To promote agricultural sustainability, the field of agriculture must be made more attractive. People must have access to the right information to ensure good agricultural practices, as well as a platform to sell their farm produce with ease encouraging people to engage in agriculture entrepreneurship. In this paper, we discussed the design and development of a mobile app (Masa) that is aimed at creating a one-stop platform where newcomers to the agricultural sector can get a first-hand walk-through guide containing all the information they need to get started and be successful in farming.

As part of our future work, we intend to evaluate the efficacy of this application among farmers to see if it has the potential to attract more people into farming, as well as its effectiveness in promoting sustainable agricultural practices. We also intend to put mechanisms in place to verify farmers who post items on the marketplace. Finally, we hope to add a feature that will allow government agencies to conveniently offer agricultural loans to registered farmers on the platform.

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